

Publications as Innovation data

Massimiliano Coda Zabetta

DAISY Summer School

7-11 September, 2026

Science of Science (SciSci)

The idea of having the **practice of science** itself as object of scientific inquiry dates back to mid-20th century → Mostly as several separate subfields within well-established disciplines.



Sociology of science

- **Merton's institutional approach:** organization of science, its institutions, communities, evaluation and reward system.

[Cole & Zuckerman, 1975; Zuckerman, 1988; Mullins, 1973; King, 1971]



Psychology of science

- Descriptive studies of the factors behind scientific discovery and of the **personal traits** linked to creativity, innovation and productivity.

[Feist, 2006; Feist & Gorman, 1998; Fisch, 1977; Simonton, 2004]



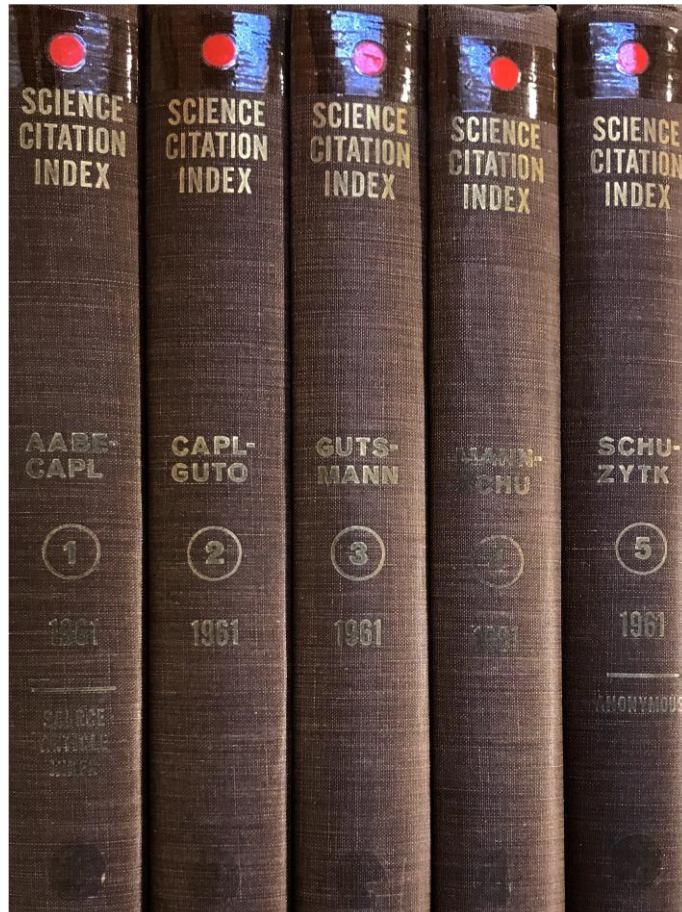
Economics of science

- Focus extended to **technology:** how science and technology affect **economic growth**, and how their study can improve policy.

[Dasgupta & David, 1994; Freeman, 1977; Stephan, 1996, 2012]

SciSci's focus has especially been on **references** (links from one document to previous ones) and **citations** (links to one document by subsequent ones), not without critiques [Hicks et al., 2015].

ISI's Science Citation Index (E. Garfield, 1963)



ABEL 139

ABELL MR-----	*50*	ARCH PATHOL-----	50	1
EMERY GN		CAN J BIOCH	61	39 977
-----	50-	ARCH PATH-----	50	23
HRSTKA V		ARCH I PHAR	61	130 304
-----	56-	ARCH PATH-----	61	61 360
WILLIAMS GE		J PATH BACT	61	82 281
-----	57-	AMER J CLIN PATH----	28	272
INKLEY SR		ARCH IN MED	61	108 903
LAUFER A		PATH MICROB	61	24 72
-----	61-	CANCER-----	14	318
GOSLING JR		CANCER	61	14 330

The quantitative turn

The emergence of SciSci has been driven by two key factors [Fortunato et al., 2018]:

- 1. Data availability.** Alongside proprietary databases (Web of Science, Scopus), open sources such as PubMed, OpenAlex and PatentsView now provide millions of records on researchers, publications and patents, covering every country and scientific field.
- 2. Transdisciplinarity.** Natural, computational and social scientists have moved into the field, bringing large-scale data skills that make it possible to test generative models of how science, its institutions and its workforce evolve.

Scientific Publications as Data

Scientific Publications as Data

Bibliographical information

- Publisher
- Journal
- Volume
- Issue
- ISBN
- ISSN

Research Policy 44 (2015) 833–848



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Knowledge flows and the absorptive capacity of regions

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ARTICLE INFO

Article history:
Received 14 January 2014
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Keywords:
Absorptive capacity
Inventor mobility
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ABSTRACT

This paper assesses the extent to which absorptive capacity determines knowledge flows' impact on regional innovation. In particular, it looks at how regions with large absorptive capacity make the most of external inflows of knowledge and information brought in by means of inventor mobility and networks, and fosters local innovation. The paper uses an unbalanced panel of 274 regions over 8 years to estimate a regional knowledge production function with fixed-effects. It finds evidence that inflows of inventors are critical for wealthier regions, while it has more nuanced effects for less developed areas. It also shows that regions' absorptive capacity critically adds a premium to tap into remote knowledge pools conveyed by mobility and networks.

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Scientific Publications as Data

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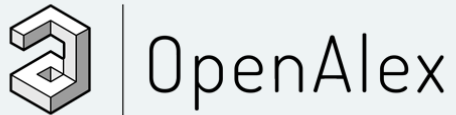
ABSTRACT

This paper assesses the extent to which absorptive capacity determines knowledge flows' impact on regional innovation. In particular, it looks at how regions with large absorptive capacity make the most of external inflows of knowledge and information brought in by means of inventor mobility and networks, and fosters local innovation. The paper uses an unbalanced panel of 274 regions over 8 years to estimate a regional knowledge production function with fixed-effects. It finds evidence that inflows of inventors are critical for wealthier regions, while it has more nuanced effects for less developed areas. It also shows that regions' absorptive capacity critically adds a premium to tap into remote knowledge pools conveyed by mobility and networks.

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Data sources

Data sources



Open (CC0)

- Open research graph
- Free API without a key (rate-limited) and full data snapshots.
- Best for large-scale, reproducible analysis.

320M+

works indexed



Licensed

- Curated citation index
- Web access and APIs require an institutional subscription.
- Best for evaluation, historical continuity and curated workflows.

97M+

Core Collection records



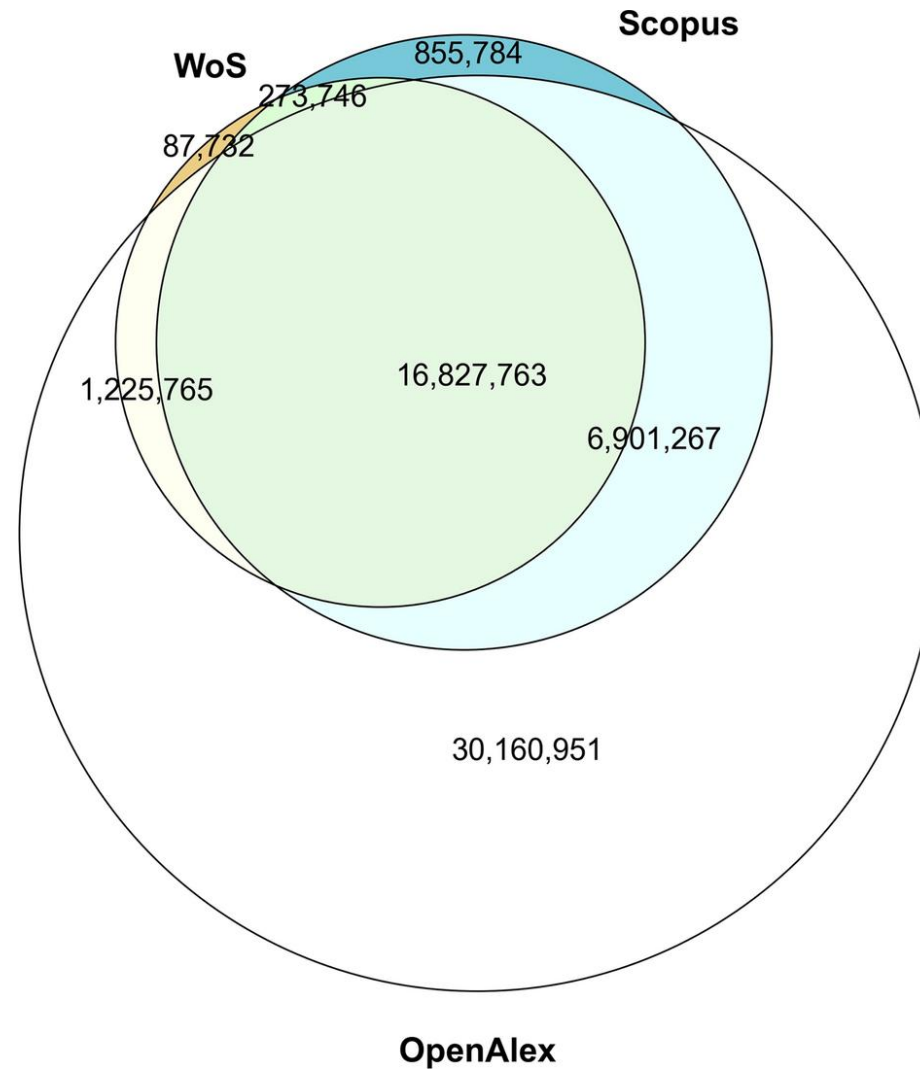
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- Best for evaluation, historical continuity and curated workflows.

100M+

records

Data sources



Data sources



Open, free

- Biomedical index with MeSH-based search.
- Free website, free API (higher limit with a free key) and downloadable yearly baseline.
- Best for biomedical analysis.

40M+

records



Mixed

- Linked research information system
- Free web app for personal use; API and analytics are paid.
- Cross-domain links pubs with grants and patents.

164M+

publications, linked to 150M patents and 7M grants



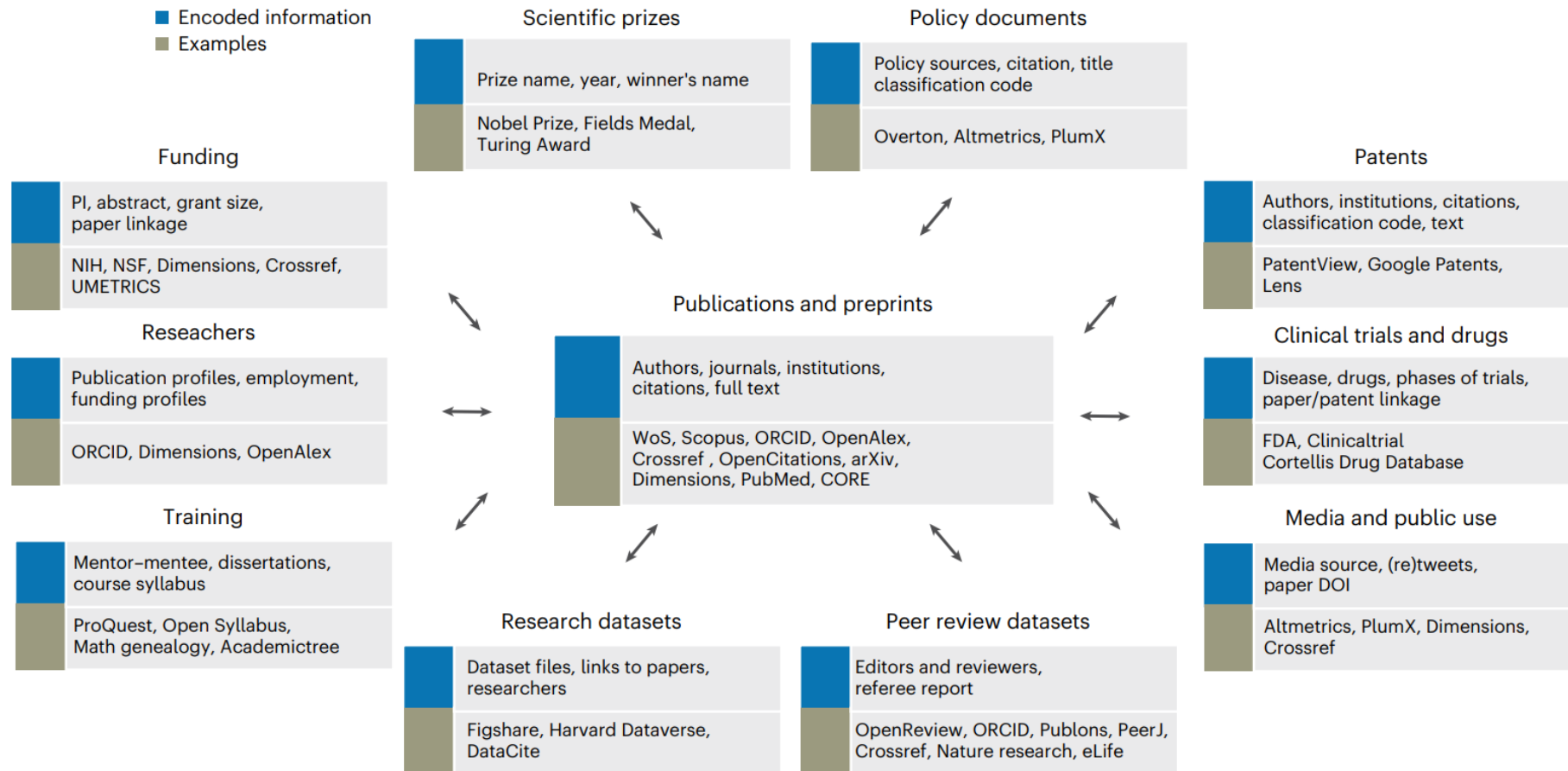
Open

- DOI registry metadata.
- Free REST API without a key (rate limits apply) plus a yearly public data file; paid Metadata Plus adds guarantees.
- Best for identifier joins.

185M+

metadata records

Data sources



Source: [Liu, L., Jones, B.F., Uzzi, B. et al., Nat Hum Behav, 2023](#)

Patent citations



Patent citations

- Citations to previous **patents** (*patent*→*patent*)
- Citations to other documents (*patent*→*non-patent reference*)
 - Citations to **scientific publications** (*patent*→*paper*)
 - Citations to **other** non-patent documents (e.g., technical reports, conference proceedings, textbooks etc.)
- Used to
 - Estimate **impact** (or market value) of patents (*patent*→*patent*)
 - Track **knowledge flows** and **spillovers** (*patent*→*patents* and *patent*→*paper*)

In-text citations

(12)	United States Patent Subramanian et al.	(10) Patent No.: US 8,282,728 B2 (45) Date of Patent: Oct. 9, 2012
(54)	MATERIALS WITH TRIGONAL BIPYRAMIDAL COORDINATION AND METHODS OF MAKING THE SAME	6,541,112 B1 4/2003 Swiler et al. 6,541,645 B1* 4/2003 Canary et al. 549/5 6,582,814 B2 6/2003 Swiler et al. 7,024,068 B2* 4/2006 Canary et al. 385/15 2003/0229131 A1* 12/2003 Sessler et al. 514/410
(75)	Inventors: Munirpallam A. Subramanian , Philomath, OR (US); Arthur W. Sleight , Philomath, OR (US); Andrew E. Smith , Rice Lake, WI (US)	OTHER PUBLICATIONS Smith, Andrew E. et al., "Mn3+ in Trigonal Bipyramidal Coordination: A New Blue Chromophore" J. Am. Chem. Soc. vol. 131, No. 47 (available online on Nov. 9, 2009) pp. 17084-17086.* Subramanian, Munirpallam A. et al., "Novel tunable ferroelectric compositions: Ba1-xLnxTi1-xMxO3 (Ln=La, Sm, Gd, Dy; M=Al, Fe, Cr)" Solid State Sciences 2 (2000) pp. 507-512.*
(73)	Assignee: State of Oregon Acting by and through the State Board of Higher Education on behalf of Oregon State University , Corvallis, OR (US)	(Continued)
(*)	Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 197 days.	<i>Primary Examiner</i> — Jessica L. Ward <i>Assistant Examiner</i> — Ross J. Christie (74) <i>Attorney, Agent, or Firm</i> — Klarquist Sparkman, LLP
(21)	Appl. No.: 12/802,700	
(22)	Filed: Jun. 10, 2010	
(65)	Prior Publication Data US 2010/0317503 A1 Dec. 16, 2010	(57) ABSTRACT Embodiments of compositions comprising materials satisfying the general formula AM _{1-x} M' _x O _{3+y} are disclosed, along with methods of making the materials and compositions. In some embodiments, M and M' are +3 cations, at least a portion of the M cations and the M' cations are bound to oxygen in trigonal bipyramidal coordination, and the material is chromophoric. In some embodiments, the material forms a crystal structure having a hexagonal unit cell wherein edge a has a length of 3.50-3.70 Å and edge c has a length of 10-13 Å. In other embodiments, edge a has a length of 5.5-7.0 Å. In particular embodiments, M' is Mn, and Mn is bonded to
	Related U.S. Application Data (60) Provisional application No. 61/268,479, filed on Jun. 11, 2009. (51) Int. Cl. C04B 14/00 (2006.01) (52) U.S. Cl. 106/401 ; 106/31.13; 501/41; 501/42; 501/50; 501/53; 501/152 (58) Field of Classification Search 106/31.13,	

TABLE 8-continued

Crystal data and structure refinement YIn _{0.33} Mn _{0.66} O ₃	
Crystal system	Hexagonal
Space group	P6 ₃ cm
Unit cell dimensions	a = 6.1709(6) Å c = 11.770(2) Å
Volume	388.17(9) Å ³
Z	6
Density (calculated)	5.437 mg/m ³
Absorption coefficient	28.267 mm ⁻¹
F(000)	576
Crystal size	0.05 × 0.03 × 0.01 mm
Theta range for data collection	3.46 to 28.31°
Index ranges	-7 ≤ h ≤ 8, -7 ≤ k ≤ 7, -15 ≤ l ≤ 15
Reflections collected	3766
Independent reflections	363 [R _{int}] = 0.0263]
Completeness to theta = 28.31°	98.0%
Absorption correction	Semi-empirical from equivalents
Max. and min. transmission	0.7653 and 0.3322
Refinement method	Full-matrix least-squares on F ²
Data/restraints/parameters	363/0/31
Goodness-of-fit on F ²	1.178
Final R indices [I > 2σ(I)]	R ₁ = 0.0219, wR ₂ = 0.0407
R indices (all data)	R ₁ = 0.0288, wR ₂ = 0.0438
Largest diff. peak and hole	0.934 and -0.629 e/Å ³

TABLE 9

Atomic coordinates and equivalent isotropic displacement parameters (Å ² × 10 ³)				
	x	y	z	U(eq)
Y1	0	0	0	3(1)
Y2	1/3	2/3	0.9636(1)	13(1)
Mn/In ^a	0.3342(4)	0	0.7211(4)	6(1)
O1	0.322(4)	0	0.879(3)	21(6)
O1'	0.289(5)	0	0.893(2)	9(6)
O2	0.635(3)	0	0.061(2)	9(3)
O2'	0.655(5)	0	0.034(2)	7(6)
O3	0	0	0.202(2)	24(4)
O4	1/3	2/3	0.749(1)	13(3)

U(eq) is defined as one third of the trace of the orthogonalized U^{ij} tensor.

TABLE 11

Anisotropic displacement parameters (Å ² × 10 ³)*					
	U ¹¹	U ²²	U ³³	U ¹³	U ¹²
Y1	2(1)	2(1)	6(1)	0	0
Y2	6(1)	6(1)	28(1)	0	0
Mn/In	7(1)	5(1)	5(1)	0	-1(1)
O3	33(7)	33(7)	5(7)	0	0
O4	3(3)	3(3)	33(7)	0	0

The anisotropic displacement factor exponent takes the form: -2π²h²a²U¹¹ + ... + 2π²hka²b²U¹³

*Split atoms O1, O1', O2, and O2' were refined with isotropic displacement parameters.

15 First-Principles Calculations

First-principles calculations were performed with plane-wave density functional theory using the Vienna Ab-initio Simulation Package (VASP). (Kresse, G., and Furthmüller, J., *Phys. Rev. B* 54, 11169-11186 (1996); Kresse, G., and Joubert, D., *Phys. Rev. B* 59, 1758-1775 (1999).) Exchange and correlation effects are treated on the level of Perdew, with an on-site Coulomb repulsion U = 5.0 eV and an intra-atomic exchange splitting of 1.0-5 eV for Mn d states (Liechtenstein, A. I., Anisimov, V. I., and Zaanen, J., *Phys. Rev. B* 52, R5467-R5470 (1995).) A global antiferromagnetic ordering with ferromagnetic spin planes was adopted for the simulations. Intermediates within periodic boundary conditions were studied using the supercell approach with lattice constants taken from experimental values presented in FIG. 7. The 40-atom supercells permit concentrations of x = 0.0, 0.25, 0.5, 0.75 and 1.0 while maintaining equal numbers of In and Mn atoms in each layer. An ordering was chosen in which the minority component was maximally separated in space; thus, the possibility of In or Mn clustering was ignored.

35 All structures were initialized in the centrosymmetric P6₃/mmc space group and all atomic degrees of freedom were optimized until forces were less than 0.1 meV/Å. This strict tolerance allowed accurate study of delicate features of the atomic structure, such as tiltings of the polyhedra that are responsible for ferroelectricity. (Fennie, C. J., and Rabe, K. M., *Phys. Rev. B* 72, 100103(4) (2005).)

45 Summary of Convergence Parameters:
11.450 eV plane-wave cutoff / 33.1 Ry / 16 Å H₂

- Bryan et al. (2020): **31%** of in-text citations appear on the front page, **24%** of front page citations appear in the text!

Some online patent data sources

Verluisse et al. (2026)

Received: 15 November 2022 | Revised: 30 May 2025 | Accepted: 5 October 2025

DOI: 10.1002/smj.70027

RESEARCH ARTICLE

SMS | Strategic Management Journal



WILEY

Beyond the front page: In-text citations to patents as traces of inventor knowledge

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Abstract

Research Summary: This study introduces in-text patent-to-patent citations—references embedded in the body of patent documents—as a novel data source to

Marx & Fuegi (2020)

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DOI: 10.1002/smj.3145

RESEARCH ARTICLE



WILEY

Reliance on science: Worldwide front-page patent citations to scientific articles

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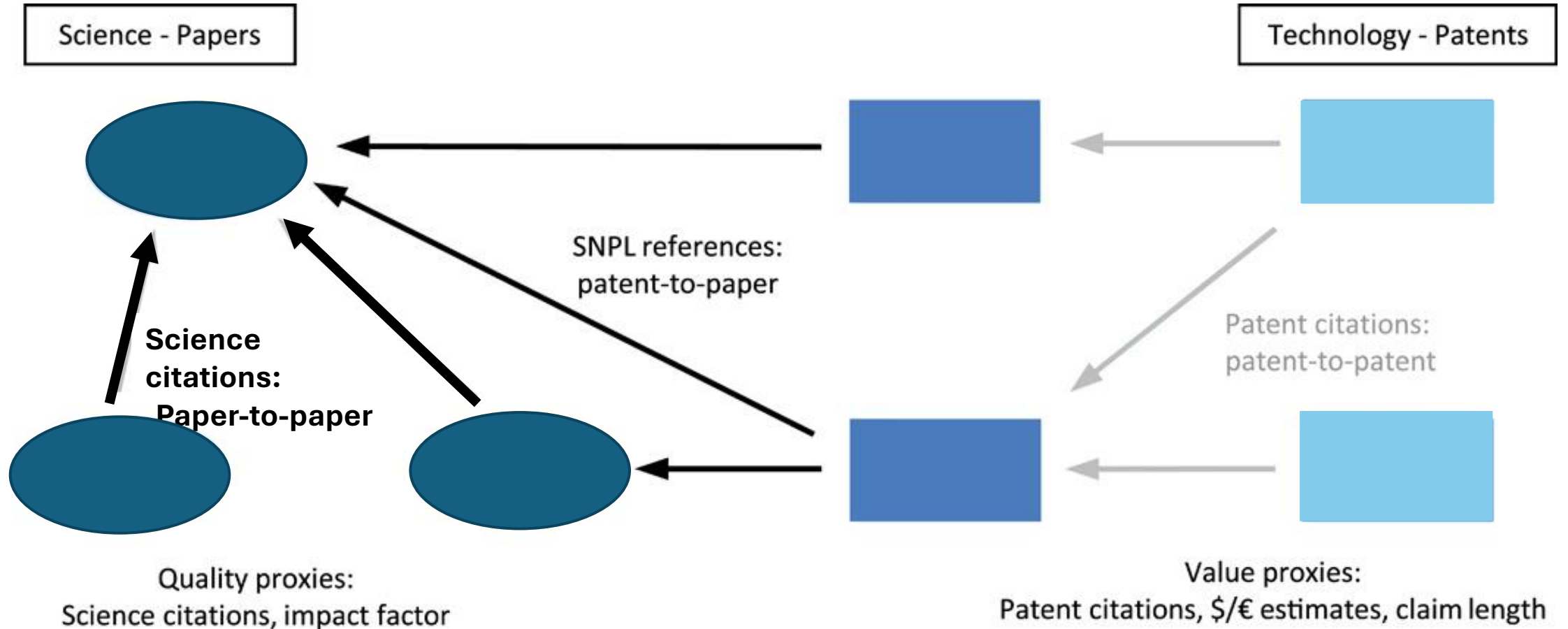
Funding information

National Science Foundation, Grant/Award Number: 1735669

Abstract

Research summary: To what extent do firms rely on basic science in their R&D efforts? Several scholars have sought to answer this and related questions, but progress has been impeded by the difficulty of matching unstructured references in patents to published papers. We introduce an open-access dataset of references from

Types of citations



OpenAlex

A fully open catalog of the global research system, named after the Library of Alexandria and run by the nonprofit OurResearch. Launched in January 2022 as the successor to **Microsoft Academic Graph** (Priem et al., 2022).

Built by aggregating Crossref and the MAG legacy, then enriched with PubMed, ORCID, ROR, DOAJ, Unpaywall and repository data. Authors, institutions and topics are disambiguated and connected algorithmically.

All data is CC0. Increasingly the open alternative for bibliometrics: adopted by institutions leaving proprietary databases (e.g. Sorbonne University) and by the Leiden Ranking Open Edition.

WHAT IT CONTAINS

320M+

works, with tens of thousands added daily

tens of millions

disambiguated author profiles

250k+

sources: journals, repositories, conferences

110k+

institutions, matched to ROR IDs

≈4,500

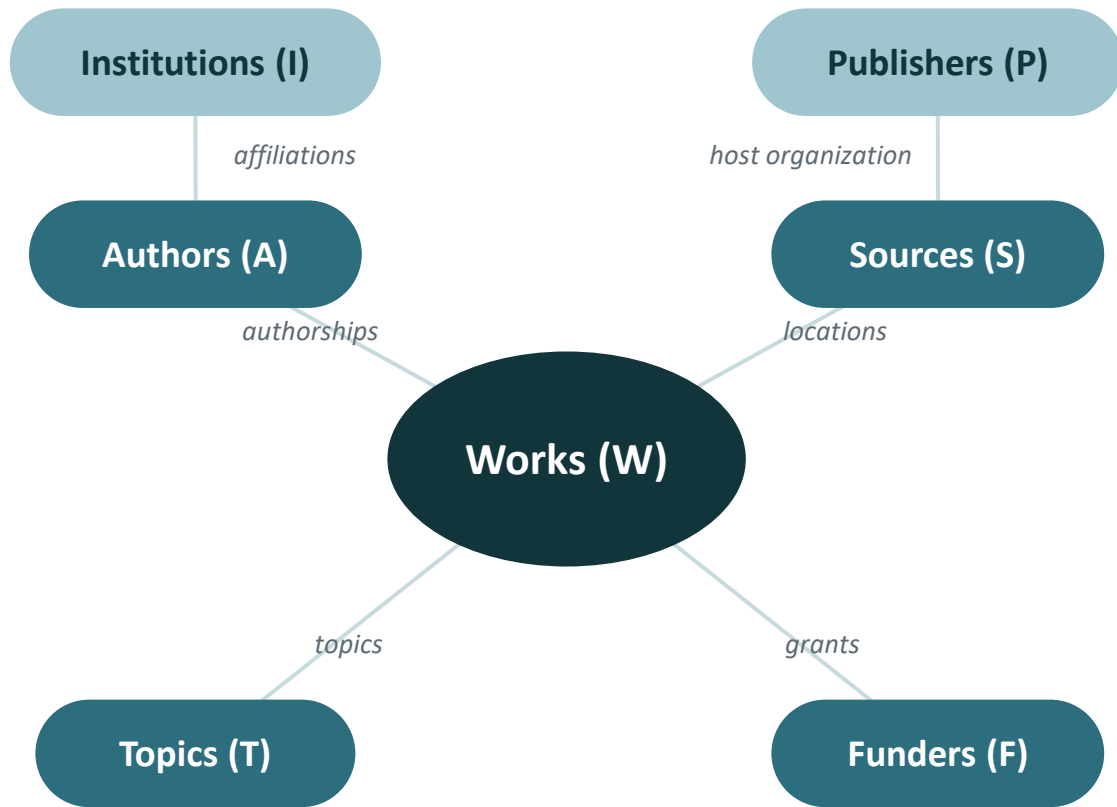
topics in a four-level hierarchy

35k+

funders, plus 10k+ publishers

Billions of connections among entities. Counts from OpenAlex documentation, mid-2026.

OpenAlex: schema and access



Works also cite works (referenced_works): the open citation graph. Keywords and geo complete the schema.

INSIDE A WORK RECORD

ids: OpenAlex ID, DOI, PMID

authorships: authors in order, with institutions and countries

locations, open_access: hosting venues and OA status

referenced_works, cited_by_count: the citation graph

topics: domain > field > subfield > topic

abstract_inverted_index: the abstract as a word-position index

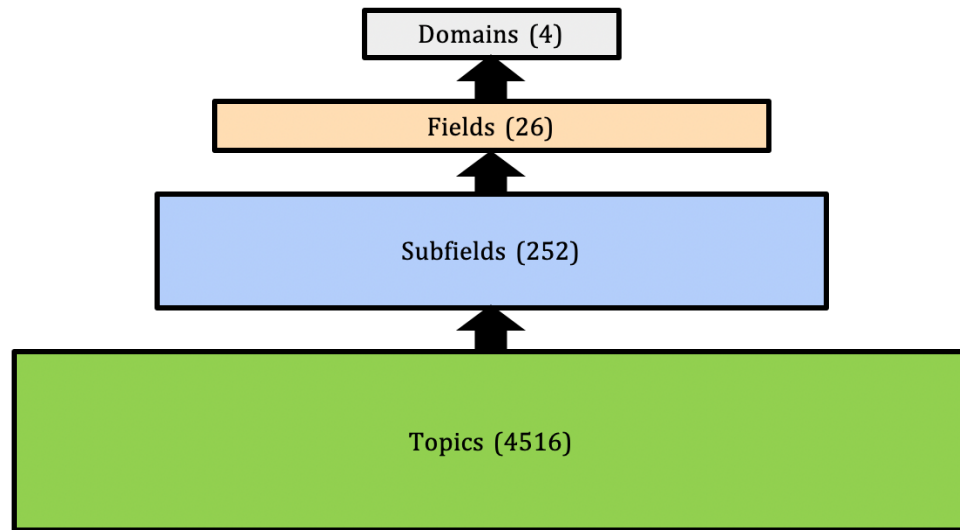
plus: type, language, grants, is_retracted, counts_by_year

HOW TO ACCESS

Web UI at openalex.org. **REST API** with a free key (required since Feb 2026), daily credit allowance, heavy bulk use metered. Monthly full **snapshot** free to download; Premium adds hourly updates and higher limits. All data CC0.

OpenAlex Topics

Topics structure



Topic example

Topic:	Natural Language Processing
Subfield:	Artificial Intelligence
Field:	Computer Science
Domain:	Physical Sciences

<https://github.com/maxcodaz/daisy-publications>